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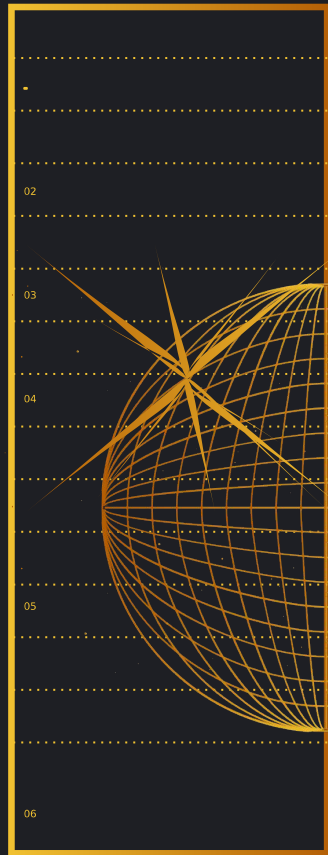
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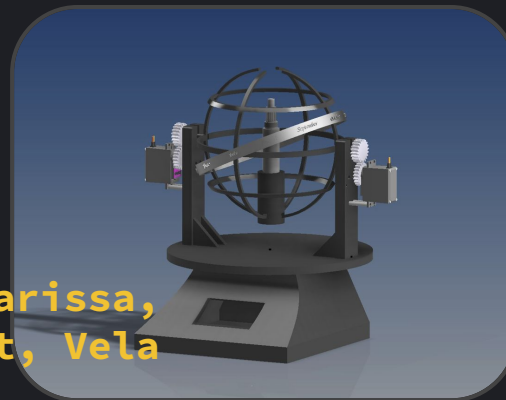


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ARMILLARY SPHERE

Group #29 - Clarissa,
Dani, Eli, Kat, Vela



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WHAT IS AN ARMILLARY SPHERE?

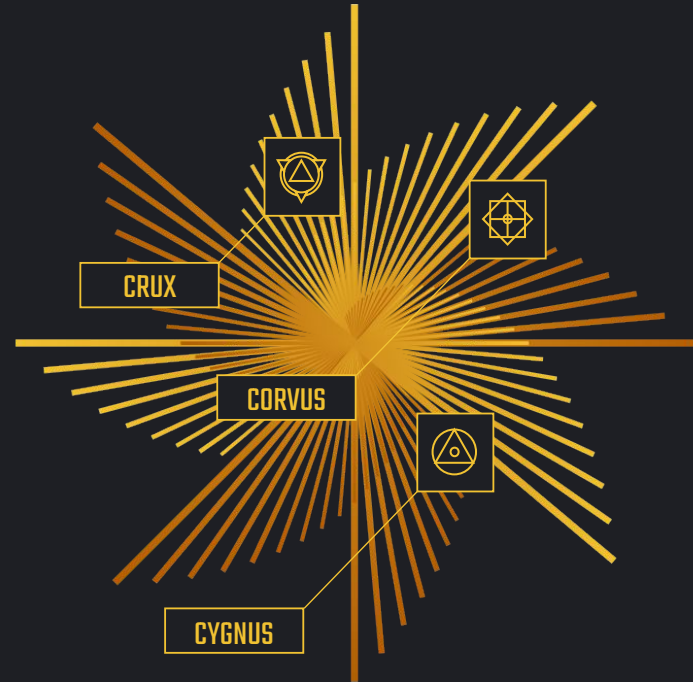


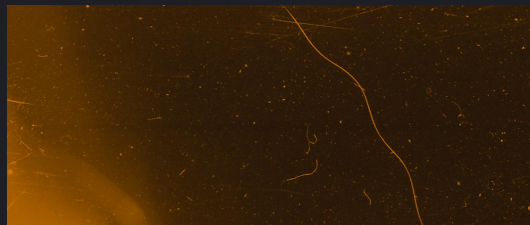
An armillary sphere is a model of the celestial sphere with **interlocking rings** that represent the Earth or Sun at the center and the **celestial equator, ecliptic, and other astronomical circles**. It was an important astronomical instrument used to demonstrate celestial motion and help astronomers **calculate positions of celestial bodies**, like the sun, moon, and planets.



OUR OBJECTIVE

Pick a celestial object you want the laser to point to, and the servo motors and stepper motor will move accordingly. The user is thus able to visualize the relative position of the celestial object, as well as view its distance.





MECHANICAL

Modeling and integration

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SOFTWARE

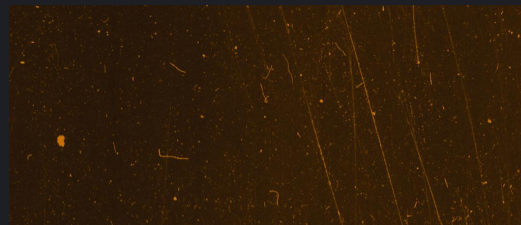
Planet and star calibration

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Controls and movement



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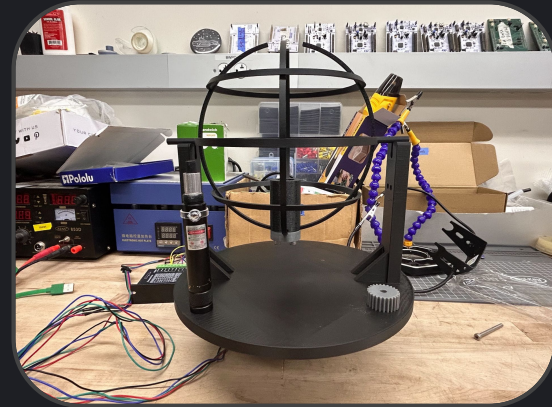
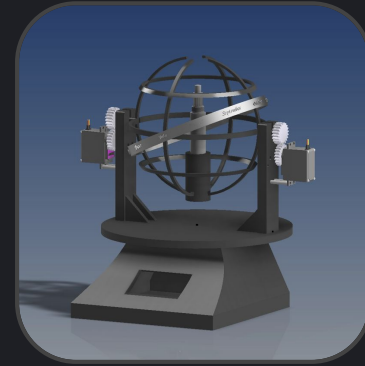
MECHANICAL

Tools: Autodesk Inventor, 3D Printers - Bambu, Filament - PLA, TPU

Challenges: 360° movement and wiring, laser mounting, ring tolerance, motor placement, gear ratio

Solution: 2 separate motors (stepper for the base, servo on the vertical stands), for its flexibility and grip - TPU filament to house the laser, several iterations and calipers to solve ring tolerance

Extra fun fact: the ecliptic ring (rotated with the months) is rotated 23.5° - to scale with the Sun's rotation around Earth!



Above: assembly process





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SOFTWARE

Tools: PySerial, Arduino IDE, Horizons API, Github

Challenges: API Calls, Data Parsing, User Geolocation, Data Output Type for API Calls

Solution: Horizons API Documentation, Creative Planning, StackOverflow, ChatGPT

```
Fetching user location...  
Location: (████████████████████)
```

```
Fetching Moon coordinates...  
Results: 1 data point(s)  
2025-Nov-29 00:00 *m 135.619130 36.882134
```

```
Fetching Moon distance...  
Results: 1 data point(s)  
2025-Nov-29 00:00 *m 0.00250713257293 -0.2767840
```

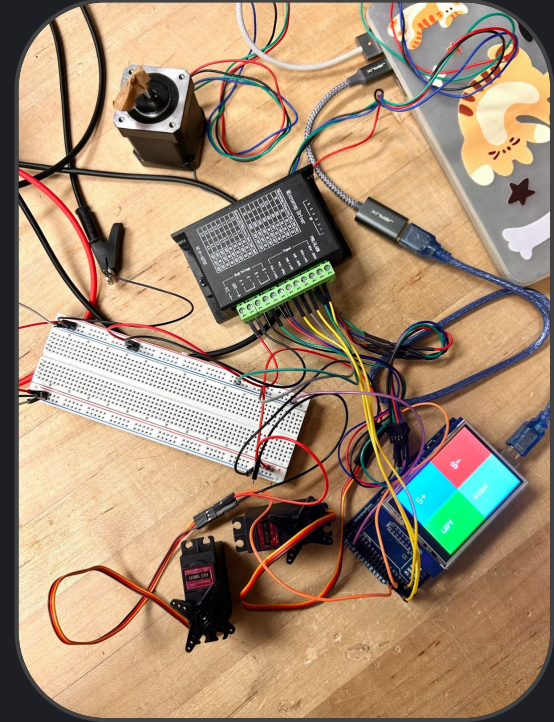


HARDWARE

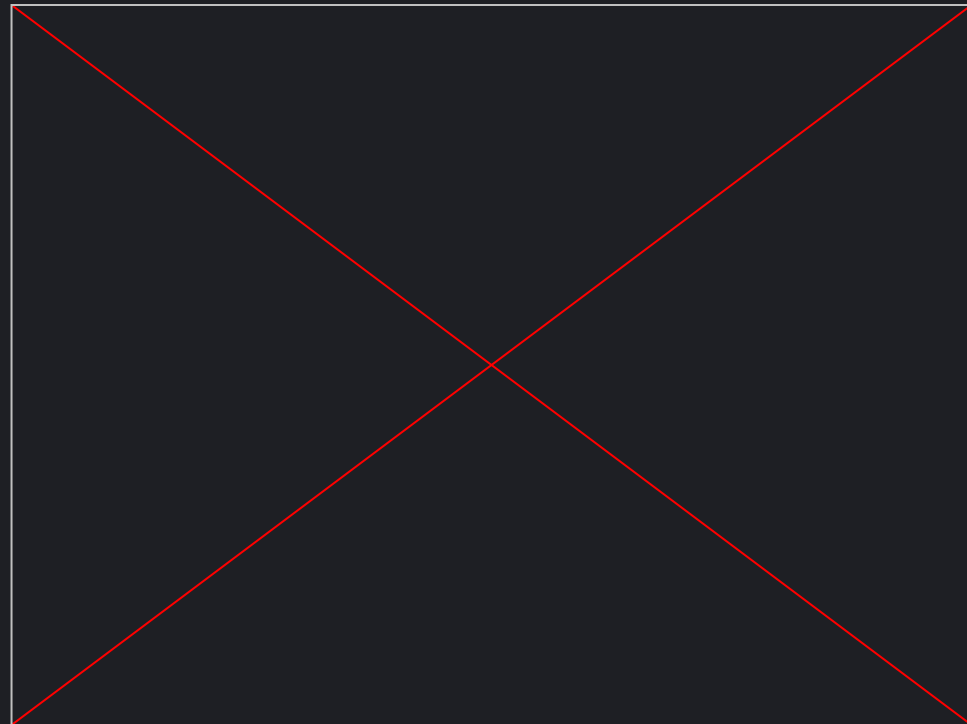
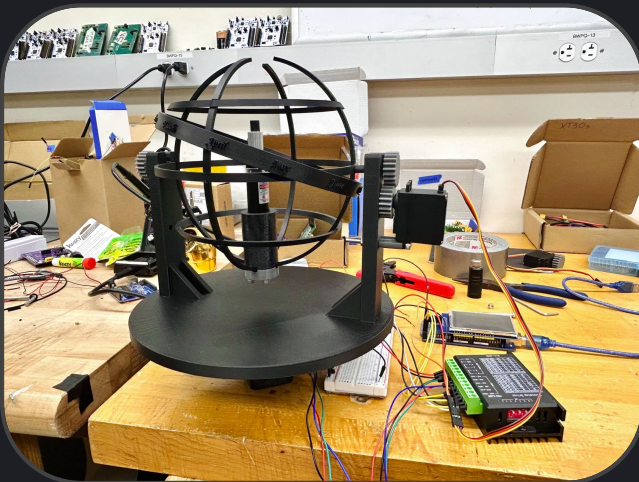
Tools: Arduino Mega, 1 Nema 17 stepper motor, 2 MG996 servo motors, TB600 motor driver, 1 LCD, breadboard, duponts, battery holders

Challenges: power distribution instability, touchscreen raw-value mapping, button areas matches to functions, serial communication delays, stepper motor driver configurations

Solution: separate batteries, full corner-based touchscreen calibration, custom mapping logic, TB600 switch configuration



Above: hardware testing



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Thank you!

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